Pharmaceutical Reform and Pharmacy Supply

County totals, hospital buffers, and within-county location

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The three outcomes answer different spatial questions

Level and outcome	Construction	Interpretation
County total	All pharmacies observed in county *c* and year *t*	Overall county pharmacy supply
Hospital buffer	Pharmacies inside 500m, 1km, or 2km of each hospital	Local environment around one hospital
County nearest distance	Each geocoded pharmacy assigned once to its nearest active public hospital	Within-county allocation: $\leq$2km versus $>$2km

- Hospital buffers may overlap, so one pharmacy can enter several hospitals' counts.
- County nearest-distance groups are mutually exclusive: each geocoded pharmacy is counted once.
- **Identity: pharmacies $\leq$2km + pharmacies $>$2km = total geocoded county pharmacies.**

The estimators target complementary reform comparisons

County-year analysis

- 2012–2019 county panel.
- County FE and province-by-year FE.
- Controls: log population and log nighttime lights.
- County-clustered standard errors.
- TWFE, CSDID, IV-DID, and event-study diagnostics.

Hospital-year analysis

- Hospital FE and province-by-year FE.
- DDD: public hospital $\times$ post reform.
- Private hospitals form the within-county comparison.
- 2km is primary; 1km and 500m are robustness radii.
- County-clustered standard errors.

Count outcomes are primary; event studies are identification diagnostics.

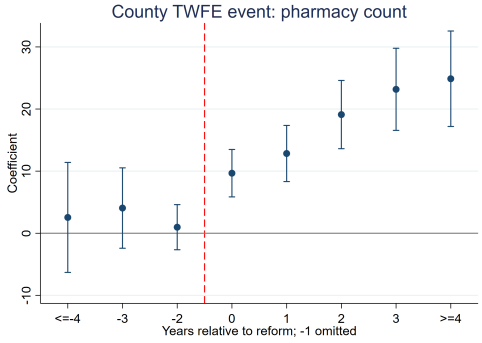
County pharmacy counts increase in both TWFE and CSDID

Estimator	Count effect	Standard error	Specification
TWFE	8.380***	(2.245)	County FE; province-by-year FE
CSDID	7.311**	(2.945)	Not-yet-treated controls

- The two estimators imply approximately **7–8 additional pharmacies per county**.
- TWFE uses 19,274 county-year observations.
- CSDID addresses staggered policy timing with group-time average treatment effects.

The county count result is the strongest and most consistent main finding.

County count dynamics support the TWFE level result

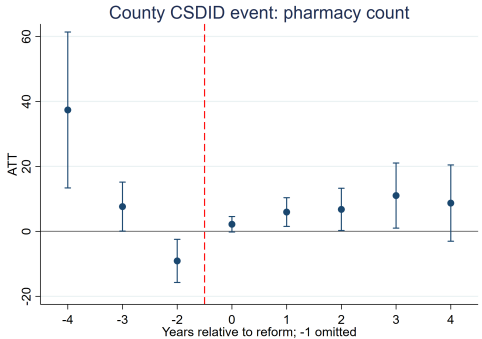


Joint pretrend test: $p = 0.669$

Relative year -1 is omitted; vertical line marks the reform threshold; 95% confidence intervals.

Post-reform coefficients rise over time without evidence of differential pretrends.

CSDID also yields a positive county count ATT



Overall ATT: 7.311 (2.945)**

Callaway–Sant'Anna CSDID with not-yet-treated controls; relative year -1 omitted.

The positive count effect survives staggered-treatment estimation, although individual pre-period estimates are uneven.

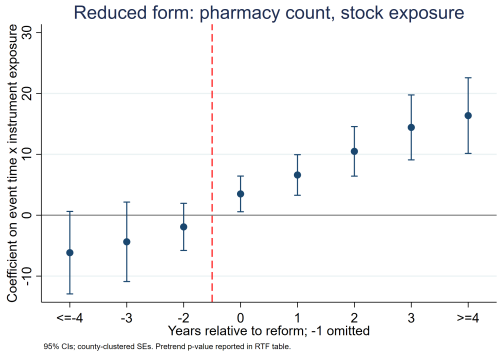
IV-DID reinforces a positive county count response

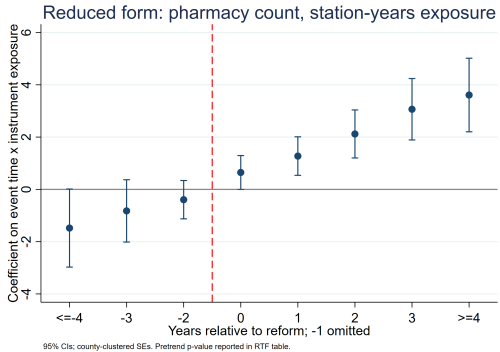
Instrument	Count IV estimate	Standard error	K-P F
1959 station stock	60.328***	(13.692)	89.719
1959 station-years	68.753***	(15.549)	76.020

- The endogenous treatment is post reform $\times$ baseline public-hospital intensity.
- Both instruments have strong first-stage diagnostics.
- The coefficient is a **continuous-intensity effect**, not the binary reform coefficient.

Static IV-DID estimates support a positive count response, but their scale is not directly comparable with TWFE or CSDID.

Dynamic IV reduced forms trend before and after reform





The dynamic IV figures are diagnostic: trending reduced forms limit a strong event-study interpretation.

Hospital buffers are local, non-unique counts

Hospital-level outcome:

$Y_{ict}(r) =$ number of pharmacies within radius r of hospital i.

Radius	Role in the paper	Spatial interpretation
2km	Primary outcome	Broad hospital-centered pharmacy market
1km	Robustness	More local hospital surroundings
500m	Robustness	Immediate walking-distance surroundings

Buffers can overlap; the same pharmacy may be counted for multiple hospitals.

The DDD coefficient compares post-reform changes near public hospitals with changes near private hospitals.

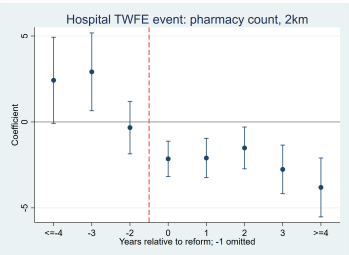
TWFE suggests fewer pharmacies near public hospitals

	2km: main	1km: robustness	500m: robustness
Public hospital × post	-3.727***	-1.184***	-0.233***
	(0.583)	(0.211)	(0.076)
Observations	138,567	138,567	138,567

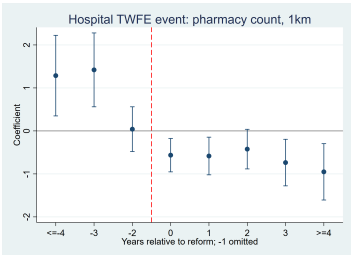
- The decline is largest at 2km and attenuates toward the hospital.
- Hospital FE and province-by-year FE absorb time-invariant hospital factors and province shocks.

Static TWFE suggests a negative hospital-centered count effect.

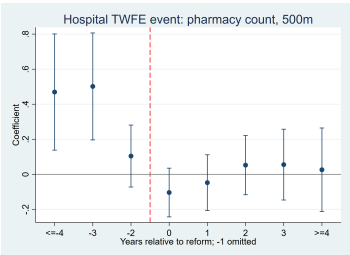
Hospital TWFE events reject parallel trends



Pretrend $p = 0.014$



Pretrend $p = 0.002$



Pretrend $p = 0.005$

All three count specifications reject parallel trends; the negative TWFE estimates are suggestive, not definitive.

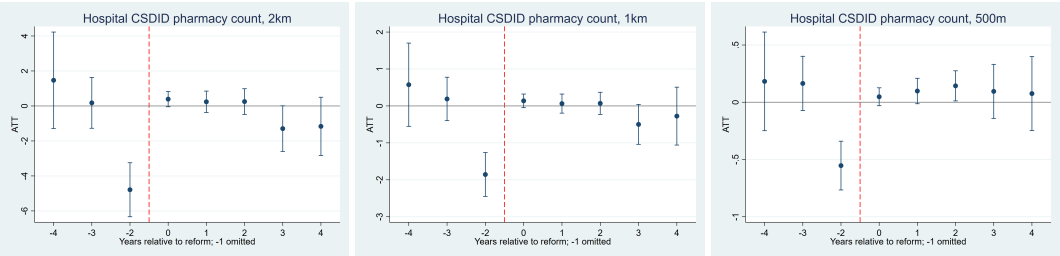
CSDID does not confirm a significant hospital count effect

	2km: main	1km: robustness	500m: robustness
Overall ATT	-0.248	-0.084	0.072
	(0.343)	(0.154)	(0.063)

- All three estimates are small and statistically insignificant.
- The 500m estimate changes sign relative to TWFE.
- The hospital count conclusion is sensitive to the staggered-treatment estimator.

CSDID does not support a robust post-reform decline around public hospitals.

Hospital CSDID dynamics are unstable



The large negative coefficient at $t = -2$ is inconsistent with a clean no-anticipation pattern.

CSDID dynamics do not rescue a causal interpretation of the hospital count result.

The county split assigns each pharmacy once

For every geocoded pharmacy-year:

$d =$ distance to the nearest active public hospital in the same county.

Group	Definition	Role
Near	Nearest public hospital $\leq$2km	Primary mechanism outcome
Far	Nearest public hospital $>$2km	Complementary outcome
500m / 1km	Within each threshold	Distance robustness only

Near + Far = total geocoded county pharmacies

Validated for 19,590 county-year rows; zero identity violations.

Static county estimates imply near gains and far losses

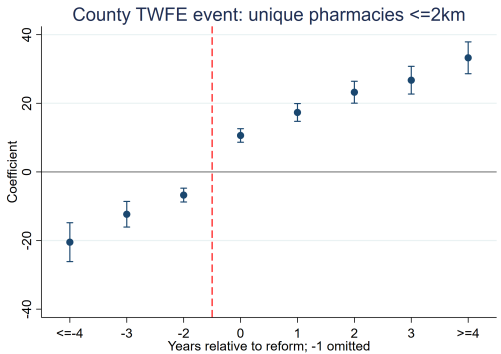
Estimator	$\leq$2km	$>$2km	Add-up to county effect
TWFE	12.498*** (1.425)	-4.118** (2.053)	$12.498 - 4.118 = 8.380$
CSDID	25.253*** (0.837)	-17.942*** (2.947)	$25.253 - 17.942 = 7.311$

Distance robustness within the near group

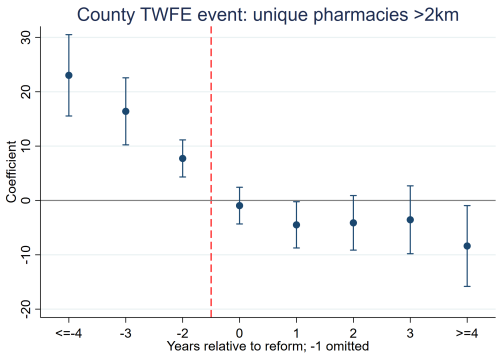
	$\leq$500m	$\leq$1km	$\leq$2km
TWFE effect	3.280*** (0.415)	7.577*** (0.867)	12.498*** (1.425)

The static decomposition is consistent with pharmacies becoming more concentrated near public hospitals.

Mechanism TWFE events show strong pretrends



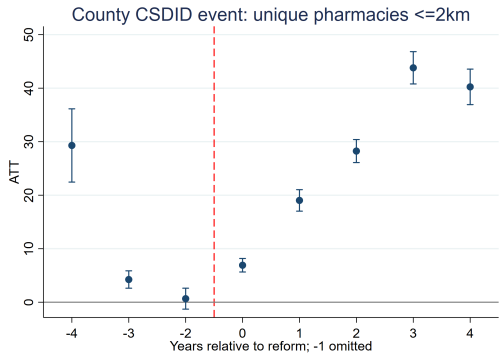
Pretrend $p < 0.001$

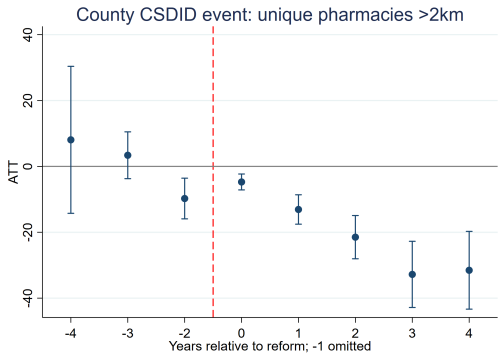


Pretrend $p < 0.001$

The decomposition is exact, but the spatial reallocation cannot be given a clean causal interpretation.

CSDID near-far divergence begins before reform





Near-pharmacy growth is visible before treatment

Far-pharmacy decline is visible before treatment

Treat the near-far split as a descriptive mechanism pattern and motivate further identification checks.

County supply evidence is strongest

- **County total count:** TWFE and CSDID both imply about 7–8 additional pharmacies; the TWFE count event study passes the pretrend test.
- **IV-DID:** Static estimates are positive with strong first stages, but dynamic reduced forms trend around the policy date.
- **Hospital buffers:** TWFE estimates are negative, but CSDID is null and count event studies reject parallel trends.
- **County near-far mechanism:** Exact add-up suggests concentration near public hospitals, yet strong pretrends prevent a causal mechanism claim.

Recommended paper hierarchy

County total count $\rightarrow$ IV-DID support $\rightarrow$ hospital 2km count $\rightarrow$ 1km/500m robustness $\rightarrow$ descriptive near-far mechanism